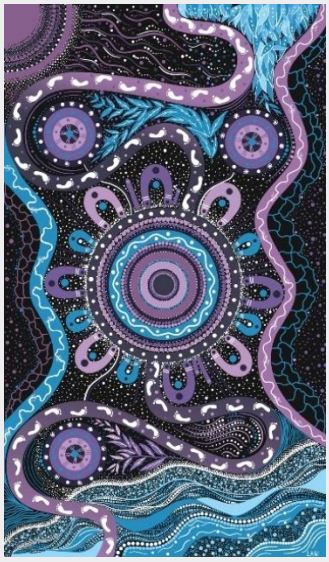
Trip of Yallourn 'W' Power Station unit 3 and unit 4 on 16 October 2025

March 2026

Reviewable Operating Incident Report under the National Electricity Rules





We acknowledge the Traditional Custodians of the land, seas and waters across Australia. We honour the wisdom of Aboriginal and Torres Strait Islander Elders past and present and embrace future generations.

We acknowledge that, wherever we work, we do so on Aboriginal and Torres Strait Islander lands. We pay respect to the world's oldest continuing culture and First Nations peoples' deep and continuing connection to Country; and hope that our work can benefit both people and Country.

'Journey of unity: AEMO's Reconciliation Path' by Lani Balzan

AEMO is proud to have launched its first Reconciliation Action Plan in May 2024. 'Journey of unity: AEMO's Reconciliation Path' was created by Wiradjuri artist Lani Balzan to visually narrate our ongoing journey towards reconciliation – a collaborative endeavour that honours First Nations cultures, fosters mutual understanding, and paves the way for a brighter, more inclusive future.

Important notice

Purpose

AEMO has prepared this report in accordance with clause 4.8.15(c) of the National Electricity Rules, using information available as at the date of publication, unless otherwise specified.

Disclaimer

To inform its review and the findings expressed in this report, AEMO has been provided with data by registered participants as to the status or response of some facilities before, during and after the reviewable incident, and has also collated information from its own observations, records and systems. Any views expressed in this report are those of AEMO unless otherwise stated, and may be based on information given to AEMO by other persons. AEMO has made reasonable efforts to ensure the quality of the information in this report but cannot guarantee its accuracy or completeness.

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Contact

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The National Electricity Market (NEM) operates on Australian Eastern Standard Time (AEST). All times in this report are in AEST.

Incident review

This reviewable operating incident[1] report is prepared in accordance with clause 4.8.15(c) of the National Electricity Rules (NER). It has been prepared using information provided by AusNet Services[2], EnergyAustralia[3] and from AEMO systems.

Table 1 Summary of event

	Details
Reviewable operating incident type	Non-credible contingency event impacting critical transmission elements.
Incident details	This report relates to a reviewable operating incident[4] that occurred at 1334 hrs on 16 October 2025 in Victoria. The incident involved the trip of the Yallourn 'W' Power Station (YWPS) unit 3 and unit 4. Each unit trip involved the disconnection of its associated 220/20 kilovolt (kV) transformer.
Incident classification	Protection/control system – cooling water isolating valve actuator control circuit was damaged, likely due to long-term flooding in the valve pit, which allowed the inadvertent remote electrical operation of the valve while the valve actuator was in a local manual operation mode.
Generation impact	A total of 597 megawatts (MW) of generation was lost as a result of this incident.
Customer load impact	No load was lost as a result of this incident.
Pre-incident conditions	Prior to the incident, YWPS unit 3 was generating 299 MW and unit 4 was generating 298 MW. There were sufficient units online in Victoria for system strength with Loy Yang A (LYA) Power Station (PS) unit 1, LYA PS unit 3, Loy Yang B (LYB) PS unit 2, YWPS unit 1, YWPS unit 3 and YWPS unit 4 online. Dartmouth PS and Newport PS were bid unavailable, which reduced the available system strength combinations. EnergyAustralia was undertaking commissioning of a new alternating current (AC) distribution supply board to replace the existing distribution board which supplies the cooling system common to YWPS unit 3 and unit 4. The YWPS unit 3 and unit 4 common cooling system is separate to that of YWPS unit 1 and unit 2.
Incident key events	On 16 October 2025 the following events occurred: 1. At 1334.29 hrs, YWPS unit 4 tripped immediately following the electrical operation of the valve actuator. 2. At 1334.32 hrs, YWPS unit 3 tripped (see Figure 1). 3. At 1350 hrs, AEMO directed Jeeralang units A2, A3 and B3 to synchronise and follow targets to manage system strength. – Jeeralang unit A2 was synchronised to the network at 1401 hrs. – Jeeralang unit A3 was synchronised to the network at 1401 hrs. – Jeeralang unit B3 was synchronised to the network at 1405 hrs. 4. At 1627 hrs, YWPS unit 3 was resynchronised to the network. 5. At 1755 hrs, AEMO cancelled the directions to Jeeralang units A2, A3 and B3 from 1800 hrs on 16 October 2025. On 17 October 2025, at 0152 hrs, YWPS unit 4 was resynchronised to the network.
Incident cause	Post-incident investigation by EnergyAustralia has concluded that the trip of YWPS unit 3 and unit 4 was caused by a loss of cooling water flow to both units. This occurred during commissioning works to replace the cooling system power supply distribution board. A cooling water isolating valve actuator operated unexpectedly, resulting in the valve closing and interrupting cooling water supply common to both units. The isolating water valve actuator control circuit, likely damaged due to long-term flooding in the pit, allowed an inadvertent remote electrical operation from the new distribution board while the valve actuator was in a local manual operation mode (electrically isolated).

[1] Reviewable operating incidents are defined by NER 4.8.15(a) and the Australian Energy Market Commission (AEMC) Reliability Panel Guidelines for Identifying Reviewable Operating Incidents.

[2] AusNet Services is the primary Victorian Declared Transmission System Operator.

[3] EnergyAustralia owns and operates Yallourn 'W' Power Station (YWPS) and Jeeralang Power Station (PS).

[4] See NER 4.8.15(a)(1)(i), as the event relates to a non-credible contingency event; and the AEMC Reliability Panel Guidelines for Identifying Reviewable Operating Incidents.

	Details
	The loss of cooling water flow resulted in the loss of condensation capability and the rapid deterioration of condenser vacuum levels. Under these conditions, generator protection operated as expected, resulting in the trip of YWPS unit 3 and unit 4.
Power system response (facilities and services)	All protection and circuit breaker (CB) operations were as expected during this incident. There was no other material impact on the broader power system, load or generation. AEMO has reviewed YWPS unit 3 and unit 4 performance and has not identified any compliance issues.
Rectification	Immediately following the incident, EnergyAustralia disconnected the electrical supply cables to the affected valve actuator, with the disconnection implemented as a permanent measure to prevent any further electrical operation of the valve. EnergyAustralia also permanently disconnected the electrical supply cables to other similar cooling water isolating valve actuators in the affected cooling water system. EnergyAustralia intends to operate these cooling system water valves in manual operation mode from now on and only plans to operate these valves during maintenance.
Power system security	Following this non-credible contingency, the combination of generating units online in Victoria was below the minimum combination threshold as specified in AEMO's Victorian system strength limit advice. Based on the availability of fast-start units at the time, and accounting for the cost of direction(s), AEMO determined that three Jeeralang A or B units were required online to manage system strength in Victoria. AEMO directed Jeeralang units A2, A3 and B3 online to meet one of the secure unit combinations specified in AEMO's Victorian system strength limit advice. Jeeralang units A2 and A3 were synchronised 27 minutes after the non-credible contingency, and Jeeralang unit B3 was synchronised 31 minutes after the non-credible contingency. Prior to the incident, AEMO was aware that, to simplify the unit combination definitions for secure system strength levels, the published Victorian system strength limit advice treats all LYA and LYB units equivalently. However, there are differences in individual LYA and LYB units' system strength contributions due to variations in their machine characteristics, particularly the subtransient reactance[5] parameter. To account for this and ensure sufficient system strength is always available, in the definition of the limit advice, it is conservatively assumed that the specific Loy Yang units offering the lowest system strength contribution are dispatched first. AEMO's post-incident investigation concluded that Victoria's minimum fault level requirements were met throughout this incident[6]. Electromagnetic transient (EMT) studies further confirmed that the power system was operating in a secure state within 30 minutes following the non-credible trip of YWPS unit 3 and unit 4 and post-synchronisation of Jeeralang units A2 and A3. This was due to the favourable system strength contribution from LYA and LYB units which were online at the time. AEMO's post-incident power system security assessment further confirmed for this event that if a different combination of three LYA or LYB units were dispatched, the available fault level may have been insufficient to ensure all Victorian fault level nodes operated above their minimum requirement. The Victorian system strength limit advice is undergoing review such that additional system strength combinations can be made available for operational use. A further review is planned to identify the best approach to consider unit-specific factors while maintaining operational clarity. Following the trip of YWPS unit 3 and unit 4, the mainland frequency reached a minimum of 49.91 hertz (Hz), remaining within the normal operating frequency band (NOFB). The Frequency Operating Standard (FOS)[7] was met for this incident. AEMO will publish a report in accordance with NER 4.8.9(f) and 3.13.6A(a) in relation to the directions to Jeeralang units A2, A3 and B3.
Reclassification	AEMO assessed whether to reclassify this incident as a credible contingency event[8]. Following the incident, on 17 October 2025, EnergyAustralia advised AEMO that the trip of YWPS unit 3 and unit 4 was caused by a cooling system water valve electrical issue common to both units, the valve actuator power supply had been isolated, and the event was unlikely to re-occur. Based on this advice, AEMO was satisfied that another occurrence of this event was unlikely and did not re-classify it as a credible contingency event. AEMO appropriately applied the reclassification criteria and determined they were not met based on the information available to AEMO at the time.
Market information	AEMO issued market notices (MNs) to inform the market of the events. These MNs are summarised in Table 2. All MNs for this incident were issued in accordance with NER requirements.
Conclusions	AEMO has concluded that: 1. On 16 October 2025, YWPS unit 3 and unit 4 tripped due to the loss of cooling system water flow to both units during commissioning works to replace the cooling system power supply distribution board. The cooling water isolating valve actuator control circuit, damaged likely due to long-term flooding in the pit, allowed the inadvertent remote electrical operation of the valve while the actuator was in local manual operation mode. 2. EnergyAustralia permanently disconnected the electrical supply cables to the affected cooling water isolating valve actuator to prevent a recurrence and also disconnected the electrical supply cables to other similar cooling water isolating valve actuators in the cooling water system. These valves will now operate only in manual mode.

[5] Subtransient reactance is the initial, lowest reactance a synchronous machine presents due to a short circuit fault.

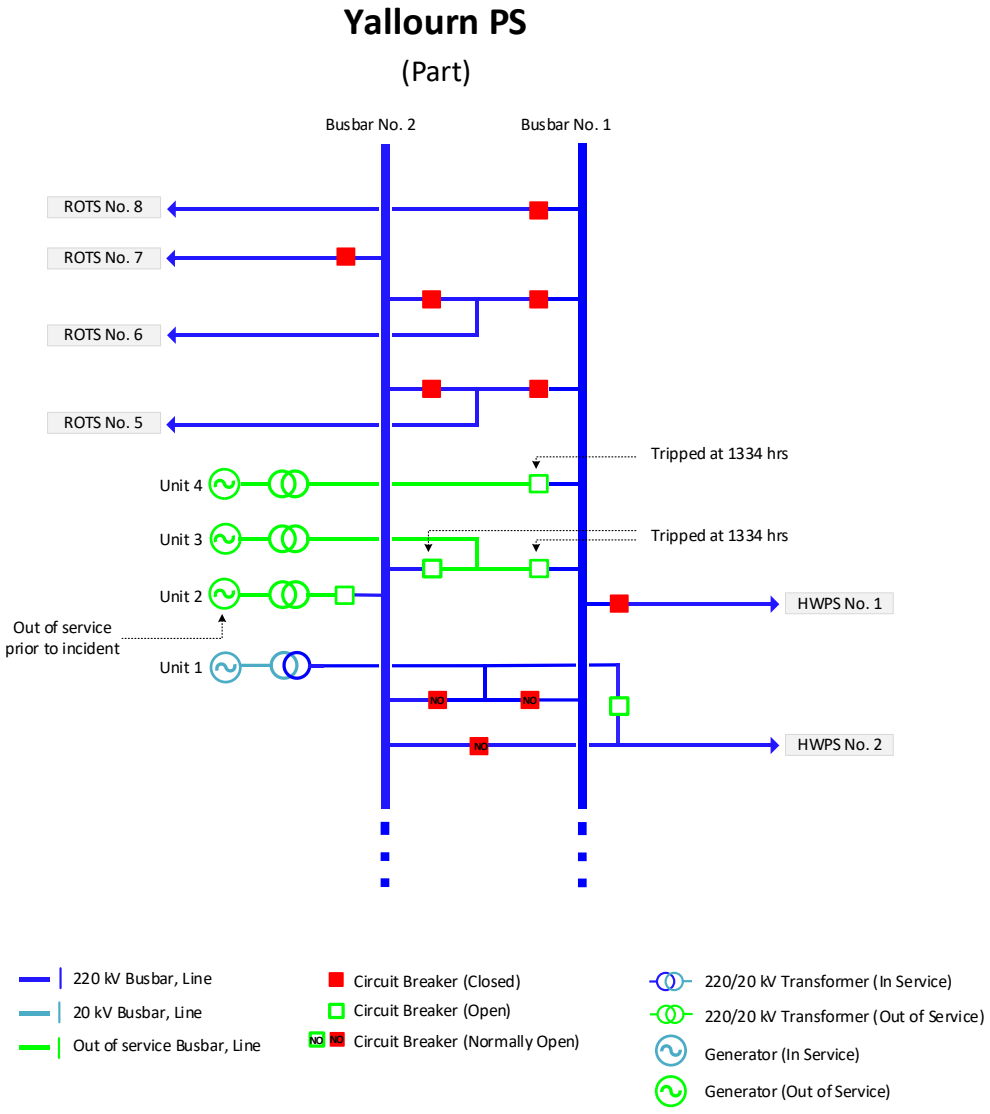
[6] See https://www.aemo.com.au/-/media/files/electricity/nem/security_and_reliability/system_security_planning/2024-system-strength-report.pdf.

[7] See https://www.aemc.gov.au/sites/default/files/2023-04/FOS%20-%20CLEAN.pdf.

[8] AEMO is required to assess whether or not to reclassify a non-credible contingency event as a credible contingency event – NER 4.2.3A(c) – and to report how the reclassification criteria were applied – NER 4.8.15(ca).

	Details
	3. Following the non-credible contingency, the Victorian system strength requirements were not met based on the Victorian system strength limit advice. AEMO directed Jeeralang units A2, A3 and B3 to synchronise and follow targets to meet one of the secure unit combinations specified in the limit advice. Jeeralang units A2 and A3 were synchronised 27 minutes after the non-credible contingency, and Jeeralang unit B3 was synchronised 31 minutes after the non-credible contingency. 4. AEMO's post-incident power system security assessments determined that the power system was operating in a secure state within 30 minutes following the incident. This was owing to the higher system strength contribution provided by the actual LYA/LYB units online at the time, relative to those assumed to be dispatched first in the limit advice. 5. The FOS was met for this incident.
Recommendations	1. AEMO to continue to consider the risks highlighted by the loss of multiple synchronous units on system strength in the 2026 General Power System Risk Review (GPSRR). 2. AEMO to share the findings of this incident with the Power System Security Working Group (PSSWG)[9] by Q2 2026. 3. VicGrid[10] to continue its planned review to identify the best approach to consider unit-specific factors into the Victorian system strength limit advice while maintaining operational clarity.

Figure 1 Post-incident diagram for 16 October 2025



[9] PSSWG is responsible for providing technical advice to the National Electricity Market Operations Committee (NEMOC) on matters related to the real-time operation of the NEM transmission system.

[10] VicGrid is responsible for the planning and development of transmission infrastructure and renewable energy zones in Victoria.

Table 2 Market Notices issued in relation to this event

MN	Issued	Details
MN 129721	1356 hrs on 16 October 2025	AEMO advised of an AEMO intervention event commencing from trading interval ending 1355 hrs on 16 October 2025 and that it had issued a direction to a participant.
MN 129720	1358 hrs on 16 October 2025	AEMO had directed EnergyAustralia to synchronise Jeeralang unit A2 and follow targets from 1350 hrs on 16 October 2025 until further notice.
MN 129722	1402 hrs on 16 October 2025	AEMO had directed EnergyAustralia to synchronise Jeeralang unit A3 and follow targets from 1350 hrs on 16 October 2025 until further notice.
MN 129723	1402 hrs on 16 October 2025	AEMO had directed EnergyAustralia to synchronise Jeeralang unit B3 and follow targets from 1350 hrs on 16 October 2025 until further notice.
MN 129724	1405 hrs on 16 October 2025	AEMO advised that this non-credible contingency event had occurred.
MN 129726	1754 hrs on 16 October 2025	AEMO had cancelled the direction to EnergyAustralia in relation to Jeeralang unit A2 (see MN 129720) from 1800 hrs on 16 October 2025.
MN 129727	1754 hrs on 16 October 2025	AEMO had cancelled the direction to EnergyAustralia in relation to Jeeralang unit A3 (see MN 129722) from 1800 hrs on 16 October 2025.
MN 129728	1755 hrs on 16 October 2025	AEMO had cancelled the direction to EnergyAustralia in relation to Jeeralang unit B3 (see MN 129723) from 1800 hrs on 16 October 2025.
MN 129729	1755 hrs on 16 October 2025	AEMO had cancelled the AEMO intervention event and all associated directions (see MN 129721) from 1800 hrs on 16 October 2025.
MN 129737	0159 hrs on 17 October 2025	AEMO advised that the cause of this non-credible contingency event had been identified and AEMO was satisfied that another occurrence of this event is unlikely under the current circumstances.

Abbreviations

Abbreviation	Term
AC	alternating current
AEMC	Australian Energy Market Commission
AEMO	Australian Energy Market Operator
AEST	Australian Eastern Standard Time
CB	circuit breaker
EMT	electromagnetic transient
FOS	Frequency Operating Standard
GPSRR	General Power System Risk Review
Hz	hertz
kV	kilovolt/s
LYA	Loy Yang A
LYB	Loy Yang B
MN	market notice
MW	megawatt/s
NEM	National Electricity Market
NEMOC	National Electricity Market Operations Committee
NER	National Electricity Rules
NOFB	normal operating frequency band
PS	Power Station
PSSWG	Power System Security Working Group
YWPS	Yallourn 'W' Power Station